Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 12: Energy Attenuation in Waveguides

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Energy Flow in Waveguides

- Energy density per unit length

$$\langle u \rangle = \frac{1}{4}\varepsilon |\vec{E}|^2 + \frac{1}{4}\mu |\vec{H}|^2$$

TE: $\psi = H_z$.

$$\vec{E}_t = -\frac{i\omega\mu}{\gamma^2}\hat{z} \times \nabla_t \psi$$

$$\vec{H}_t = \frac{ik}{\gamma^2}\nabla_t \psi$$

$$\langle u \rangle = \frac{1}{4}\varepsilon\left(\frac{\omega^2\mu^2}{\gamma^4}|\nabla_t \psi|^2\right) + \frac{1}{4}\mu\left(|\psi|^2 + \frac{k^2}{\gamma^4}|\nabla_t \psi|^2\right)$$

Energy Flow in Waveguides

- Energy density per unit length

$$\langle u \rangle = \frac{1}{4}\varepsilon\left(\frac{\omega^2\mu^2}{\beta^4}|\nabla_t\psi|^2\right) + \frac{1}{4}\mu\left(|\psi|^2 + \frac{k^2}{\beta^4}|\nabla_t\psi|^2\right)$$

$$= \frac{1}{4}\frac{\mu}{\beta^4}\left(\omega^2\mu\varepsilon + k^2\right)|\nabla_t\psi|^2 + \frac{1}{4}\mu|\psi|^2$$

$$\int \langle u \rangle \, da = \int \left(\frac{1}{4}\frac{\mu}{\beta^4}\left(\omega^2\mu\varepsilon + k^2\right)|\nabla_t\psi|^2 + \frac{1}{4}\mu|\psi|^2\right) da$$

Green's 1st identity:

$$\int_A (\nabla_t\psi)^* \cdot (\nabla_t\psi)\, da = \oint \psi^* \frac{\partial\psi}{\partial n}\, dl - \int_A \psi^* \nabla_t^2 \psi \, da$$

Energy Flow in Waveguides

- Energy density per unit length

$$\int_A (\nabla_t \psi)^* \cdot (\nabla_t \psi)\, da = \underbrace{\oint \psi^* \frac{\partial \psi}{\partial n}\, dl}_{0} - \int_A \psi^* \nabla_t^2 \psi\, da$$

$$= -\int \psi^* \nabla_t^2 \psi\, da \qquad \nabla_t^2 \psi + \gamma^2 \psi = 0$$

$$= \gamma^2 \int \psi \psi^*\, da = \gamma^2 \int |\psi|^2\, da$$

Energy Flow in Waveguides

- Energy density per unit length

$$\int \langle u \rangle \, da = \int \left(\frac{1}{4} \frac{u}{\rho^4} \left(\omega^2 u \xi + k^2 \right) |\nabla_t \psi|^2 + \frac{1}{4} u |\psi|^2 \right) da$$

$$= \int \left(\frac{1}{4} \frac{u}{\rho^4} \left(\omega^2 u \xi + k^2 \right) \rho^2 |\psi|^2 + \frac{1}{4} u |\psi|^2 \right) da$$

$$= \frac{1}{4} \left(\frac{u}{\rho^2} \left(\omega^2 u \xi + k^2 \right) + u \right) \int |\psi|^2 \, da$$

$$= \frac{1}{4} \frac{u}{\rho^2} \left(\omega^2 u \xi + k^2 + \rho^2 \right) \int |\psi|^2 da$$

$$= \frac{1}{2} \frac{u}{\rho^2} \omega^2 u \xi \int |\psi|^2 da = \frac{1}{2} u \frac{\omega^2}{\omega_\lambda^2} \int |\psi|^2 da$$

Energy Flow in Waveguides

- Energy density per unit length

$$\langle U \rangle = \int_A \langle u \rangle \, da = \frac{1}{2} \left(\frac{\omega}{\omega_c} \right)^2 \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int_A |\psi|^2 \, da \qquad \begin{matrix} TM \\ TE \end{matrix}$$

Energy Flow in Waveguides

- vg and vp

$$\langle U \rangle = \int_A \langle u \rangle \, da = \frac{1}{2} \left(\frac{\omega}{\omega_\lambda} \right)^2 \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int_A |\psi|^2 \, da \qquad \begin{matrix} TM \\ TE \end{matrix}$$

$$\langle P \rangle = \frac{1}{2\sqrt{\mu\varepsilon}} \left(\frac{\omega}{\omega_\lambda} \right)^2 \left(1 - \frac{\omega_\lambda^2}{\omega^2} \right)^{1/2} \begin{Bmatrix} \varepsilon \\ \mu \end{Bmatrix} \int_A |\psi|^2 \, da$$

$$\frac{\langle P \rangle}{\langle u \rangle} = \frac{\sqrt{1 - \frac{\omega_\lambda^2}{\omega^2}}}{\sqrt{\mu\varepsilon}} \qquad k = \sqrt{\mu\varepsilon} \sqrt{\omega^2 - \omega_\lambda^2}$$

$$= \frac{k}{\omega\mu\varepsilon} = v_g \qquad \langle P \rangle = \langle U \rangle v_g$$

Energy Attenuation in Waveguides

- Good conductor

$w_{\lambda 0}$, $\gamma_{\lambda 0}$. $k_{\lambda 0}$ perfect conductor

Good conductor : $k_\lambda = k_{\lambda 0} + \partial_\lambda + i\beta_\lambda$

? ?

- Method 1: Ohmic loss estimate
- Method 2: Boundary value problems for good conductor

β_λ V ∂_λ X

∂_λ, β_λ V

M1: Ohmic loss estimate

- Ohmic loss per length

$$e^{ikz} = e^{ik_r z - k_i z}$$

$$P(z) = P_0 e^{-2\beta_\lambda z}$$

$$\frac{dP}{dz} = -2\beta_\lambda P_0 e^{-2\beta_\lambda z} = -2\beta_\lambda P$$

$$\beta_\lambda = -\frac{1}{2}\frac{dP/dz}{P}$$

dP/dz ? P ✓

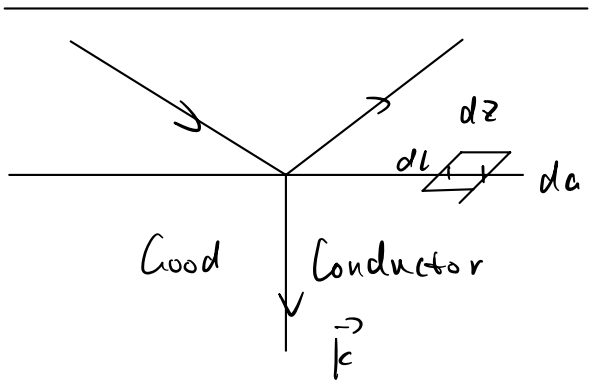
M1: Ohmic loss estimate

$da = dl\,dz$

- Ohmic loss per length

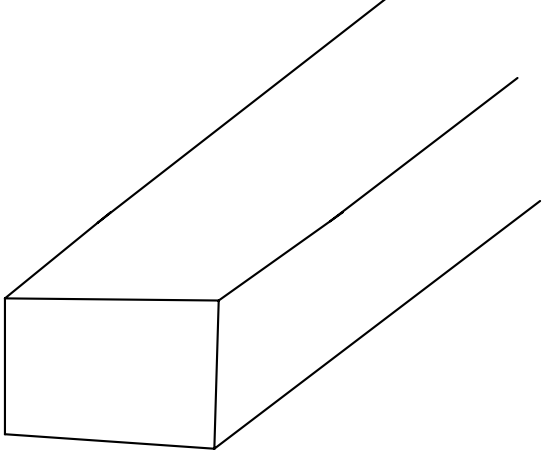
$$\frac{dP_{loss}}{da} = \frac{R_s \omega \delta}{4} \left| \vec{H}_{//} \right|^2$$

$$= -\frac{dP}{da}$$



$$\frac{dP}{dz} = \oint \frac{dP}{da}\, dl = -\oint \frac{R_s \omega \delta}{4} \left| \vec{H}_{//} \right|^2 dl$$

M1: Ohmic loss estimate



- TM mode

$$|\vec{H}_{//}| = |\vec{u} \times \vec{H}_t|$$

$$\vec{H}_t = \frac{j\beta}{m^2 i}(z_0 \nabla_t H_z \times \vec{z})$$

$$|\vec{H}_{//}| = |\vec{H}_t \times \vec{u}| = \left|\frac{j\beta}{m^2}(z_0 \nabla_t H_z \times \vec{z}) \times \vec{u}\right|$$

$$\left|\frac{\mu\varepsilon}{z_0 p}\right| \frac{j\beta}{m^2} = \frac{j\beta}{m^2}\left|\vec{z}\,(z_0 \nabla_t H_z \cdot \vec{u})\right|$$

M1: Ohmic loss estimate

- TM mode

$$-\frac{dP}{dz} = \frac{\mu_c \omega \delta}{4} \oint \frac{\xi^2 w^2}{g^4} \left| \frac{\partial \bar{b}_z}{\partial n} \right|^2 dL$$

$$= \frac{1}{2\delta\sigma} \frac{w^2}{w_\lambda^2} \oint \frac{1}{w_\lambda^2 u^2} \left| \frac{\partial \psi}{\partial n} \right|^2 dL$$

M1: Ohmic loss estimate

- TE mode

HW

$$-\frac{dP}{dz} = \frac{1}{2\delta}\left|\frac{\omega}{\omega_\lambda}\right|^2 \oint \left[\frac{1}{\varepsilon\mu\omega_\lambda^2}\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)\left|\vec{n}\times\nabla_t\psi\right|^2 + \frac{\omega_\lambda^2}{\omega^2}|\psi|^2\right] dl$$

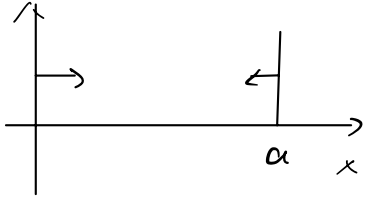
$$\beta_\lambda = -\frac{1}{2}\,\frac{dP/dz}{P}$$

Example: Rectangular waveguides

TE : $\psi_{mn} = H_0 \cos\left(\frac{m\pi x}{a}\right) \cos\left(\frac{n\pi y}{b}\right)$

$x = 0, a, \quad \hat{n} = \pm \hat{x}$

$$\left|\hat{n} \times \nabla_t H_z\right|^2 = \left(H_0 \frac{n\pi}{b} \sin\frac{n\pi y}{b}\right)^2$$



$y = 0, b, \quad \hat{n} = \pm \hat{y}$

$$\left|\hat{n} \times \nabla_t H_z\right|^2 = \left(H_0 \frac{m\pi}{a} \sin\frac{m\pi x}{a}\right)^2$$

$$\omega_\lambda^2 = \frac{\bar{\pi}^2}{\mu\varepsilon}\left(\frac{m^2}{a^2} + \frac{n^2}{b^2}\right)$$

Example: Rectangular waveguides

$$-\frac{dP}{dz} = \frac{1}{2\delta_b}\frac{\omega^2}{\omega_\lambda^2}\left\{\frac{1}{\mu\varepsilon\omega_\lambda^2}\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)\left(|A_0|^2\frac{n^2\bar{\pi}^2}{b} + |A_0|^2\frac{m^2\bar{\pi}^2}{a}\right)\right.$$

$$\left. + \frac{\omega_\lambda^2}{\omega^2}|A_0|^2 C(a,b)\right\}$$

$$C(a,b) = \begin{cases} a+b & m>0,\ n>0 \\ 2a+b & m=0 \\ a+2b & n=0 \end{cases}$$

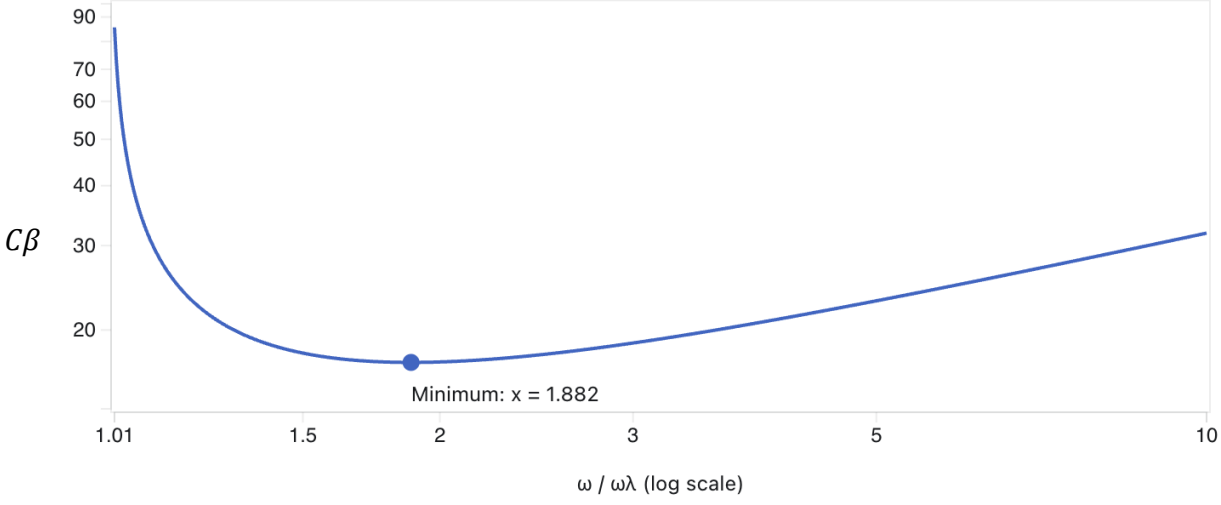
Example: Rectangular waveguides

$$P = \frac{1}{2\sqrt{\mu\varepsilon}} \frac{\omega^2}{\omega_\lambda^2} \left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)^{1/2} |A_0|^2 \begin{cases} \frac{ab}{4} & m>0,\ n>0 \\ \frac{ab}{2} & m=0 \text{ or } n=0 \end{cases}$$

$m=1,\ n=0,$

$$\beta = \frac{\sqrt{\frac{\mu}{\varepsilon}} \frac{1}{2b} \left[a\left(1 - \frac{\pi^2}{a^2\varepsilon\mu\omega^2}\right) + \frac{\omega_\lambda^2}{\omega^2}(a+2b) \right]}{ab\left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)^{1/2}}$$

Example: Rectangular waveguides



General form

General form

